Software Requirements Specification (SRS)

Restaurant Management System

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1. Introduction

1.1 Purpose

This document specifies requirements for a Restaurant Management System that integrates table reservations with pre-ordering, online food ordering, and event catering management into a unified web-based platform.

1.2 Scope

The system provides customers with convenient access to restaurant services while enabling staff to efficiently manage operations across three interdependent modules: reservation management, order processing, and event booking. The system will provide a responsive user interface accessible on both desktop and mobile devices, ensuring a consistent experience across platforms.

2. Problem Statement

Restaurants face operational inefficiencies from fragmented service delivery systems. Table reservations, online ordering, and event bookings operate independently, requiring customers to use multiple channels and staff to coordinate across disconnected platforms. This fragmentation leads to:

- Double-booking risks during peak hours due to manual reservation coordination
- Customer confusion about service availability and ordering processes
- Staff time wasted on phone calls and manual data entry across systems
- Resource conflicts between regular operations and event bookings

An integrated system is needed to unify customer interactions and operational management, eliminating coordination overhead while improving service quality.

3. Objectives and Success Criteria

Success will be measured through specific, testable criteria:

3.1 Performance

- All user operations complete within 3 seconds under normal load (50 concurrent users)
- Database queries execute within 1 second for availability checks and order retrieval

3.2 Availability

- System maintains 95% uptime during operational hours (10 AM - 11 PM)
- Zero double-booking incidents through proper database constraints

3.3 Security

- 100% enforcement of role-based access control across all user types
- User sessions timeout after 15 minutes of inactivity

4. Stakeholders

- Restaurant Owners: Require operational efficiency, revenue optimization, and business insights
- Customers: Need convenient access to reservations, ordering, and event booking
- Restaurant Staff: Manage daily operations including reservation approval and order processing
- Kitchen Staff: Receive and process orders with proper preparation timelines
- Event Coordinators: Handle catering requests and resource allocation for special events

5. Complete System Functions

A comprehensive restaurant management system should include:

- User Management: Registration, authentication, profile management, role-based permissions
- Table Reservation: Availability search, booking creation, modification, cancellation, preorder integration
- Online Ordering: Menu browsing, cart management, payment processing, order tracking, delivery/takeaway
- Event Management: Event booking requests, space allocation, custom menu selection, approval workflows
- Menu Management: Item creation, categorization, pricing, availability updates
- Reporting: Basic operational reporting of reservations and orders.
- Staff Management: Role assignment, dashboard access, task workflows

6. Course Project Implementation Scope

The prototype will implement three core interdependent modules with role-based access control:

6.1 Module 1: Table Reservation and Pre-Order

Customer Features:

- User registration and authentication
- Table availability search by date, time, and party size
- Reservation creation with automatic table assignment
- Pre-order menu browsing and item selection during reservation
- View and cancel upcoming reservations

Staff Features:

- Reservation approval queue and review
- Approve or reject reservations
- Table status dashboard and management
- Pre-order viewing for kitchen preparation planning

6.2 Module 2: Online Ordering and Tracking

Customer Features:

- Menu browsing with category organization
- Shopping cart with add, remove, and quantity management
- Order type selection (delivery or takeaway)
- Delivery address entry and payment method selection

- Order placement with confirmation and tracking
- Order history with reorder functionality

Staff Features:

- Order queue management with status filtering
- Order status updates (Received → Preparing → Ready → Completed)
- Order search by order number or customer

6.3 Module 3: Event and Catering Booking

Customer Features:

- Event booking form with type, date, time, and guest count
- Event space availability checking
- Catering menu selection with per-person pricing
- Special requirements entry and booking submission

Staff Features:

- Event booking request queue and review

6.4 Cross-Module Features

- Two user roles: Customer, Staff
- Role-specific dashboards and navigation
- Menu management (create, edit, delete items)
- Responsive web interface supporting mobile to desktop

7. Functional Requirements

7.1 User Authentication

- The system shall allow customer registration with email, password, name, and phone number
- The system shall authenticate users and create sessions upon successful login
- The system shall lock accounts for 15 minutes after three failed login attempts

7.2 Table Reservation

- The system shall display available time slots based on date, time, and party size
- The system shall automatically assign tables based on party size and capacity
- The system shall allow pre-order selection during reservation with special instructions
- The system shall create reservations with Pending Approval status requiring staff confirmation
- The system shall allow customers to cancel reservations at least 24 hours in advance

7.3 Online Ordering

- The system shall display menu items organized by categories with details and pricing
- The system shall maintain shopping cart state during user sessions
- The system shall validate delivery addresses against configured service zones
- The system shall process orders and generate unique order numbers with confirmation
- The system shall track order status through Received, Preparing, Ready, and Completed stages

7.4 Event Booking

- The system shall check event space availability based on date, time, and guest capacity
- The system shall calculate catering costs based on guest count and selected menu items
- The system shall prevent double-booking of event spaces through conflict detection
- The system shall allow staff to approve or reject event requests with customer notifications

7.5 Administrative Functions

- The system shall allow administrators to create, edit, and delete menu items and categories
- The system shall allow table and event space configuration with capacity settings
- The system shall allow administrators to create staff accounts with role assignment

8. Non-Functional Requirements

8.1 Performance Requirements

- Response Time: All user operations shall complete within 3 seconds under normal load
- Concurrent Users: System shall support 50 concurrent users without degradation
- Database Performance: Availability and order queries shall execute within 1 second

8.2 Availability Requirements

- System Uptime: Platform shall maintain 95% availability during operational hours (10 AM - 11 PM)
- Data Integrity: Zero double-booking incidents through database constraints and transaction management
- Recovery: System shall recover from failures within 15 minutes

8.3 Security Requirements

- Authentication: All protected features require valid user authentication
- Authorization: Role-based access control prevents users from accessing unauthorized functions
- Data Protection: Passwords stored using secure hashing, sensitive data encrypted in transmission
- Session Security: Sessions timeout after 15 minutes of inactivity

8.4 Usability Requirements

- Responsive Design: Interface shall function on devices from 320px (mobile) to 1920px (desktop) width
- Task Efficiency: Primary workflows complete within 5 user interactions
- The interface shall be optimized for mobile devices, supporting touch interactions, smaller screen layouts, and adaptive content display.
- Error Handling: Clear error messages with recovery paths for all failure scenarios

8.5 Scalability Requirements

- User Load Scaling: The system shall be capable of handling up to 200 concurrent users without performance degradation and should allow horizontal scaling to support future growth.

9. System Constraints

- Business Hours: System operates according to configured hours (typically 10 AM - 11 PM)
- Advance Booking: Reservations accepted up to 30 days ahead, minimum 2 hours before slot
- Cancellation Policy: Reservations cancelled only 24+ hours before scheduled time
- Delivery Zone: Online orders limited to configured service areas
- Payment: Prototype simulates payment processing without actual gateway integration
- Web Interface: Accessed via modern browsers with JavaScript enabled, no offline support

10. Key User Workflows

10.1 Customer Makes Reservation with Pre-Order

1. Customer logs in and navigates to reservation page
2. Customer selects date, time, and party size, then searches availability
3. System displays available time slots with automatic table assignment
4. Customer selects time slot and optionally adds pre-order items from menu
5. Customer confirms reservation; system creates pending reservation with notification
6. Staff reviews request and approves; system sends confirmation to customer

10.2 Customer Places Online Order

1. Customer browses menu by category and adds items to cart
2. Customer reviews cart and proceeds to checkout
3. Customer selects delivery or takeaway, enters address/pickup time
4. Customer chooses payment method and confirms order
5. System creates order with unique number and sends confirmation
6. Staff processes order, updating status from Received → Preparing → Ready → Completed

10.3 Customer Books Event

1. Customer enters event details (type, date, time, guest count, duration)
2. System checks event space availability and displays suitable spaces
3. Customer selects space and browses catering menu
4. Customer selects menu items, adds special requirements, and submits request

5. System creates pending event booking and notifies staff
6. Staff reviews for conflicts, approves booking, and confirms with customer